Volatility Arbitrage via the Variance Risk Premium

Trading the IV–Forecast RV Spread on SPX Options

Group 4
MS&E 244: Statistical Arbitrage

Stanford University

Winter 2026

Motivation & Strategy

Motivation

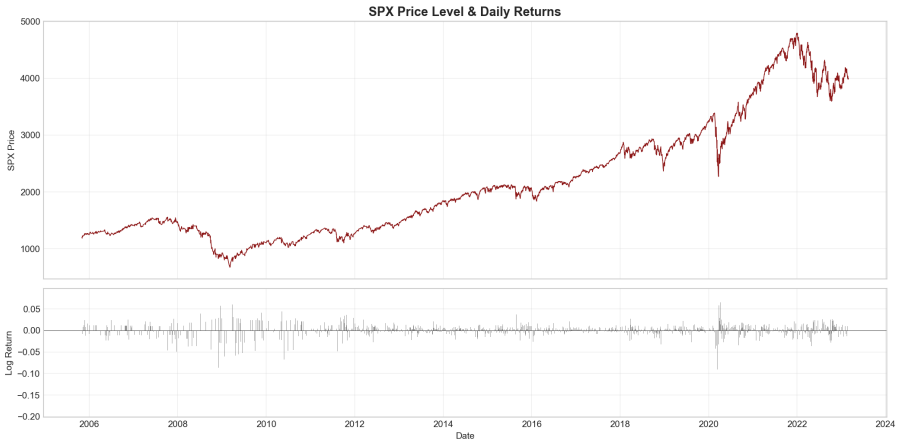
- **VRP:** Implied variance > realized variance.
- Compensates sellers for crash risk.
- **Question:** Is IV–forecast RV spread tradeable?

Strategy

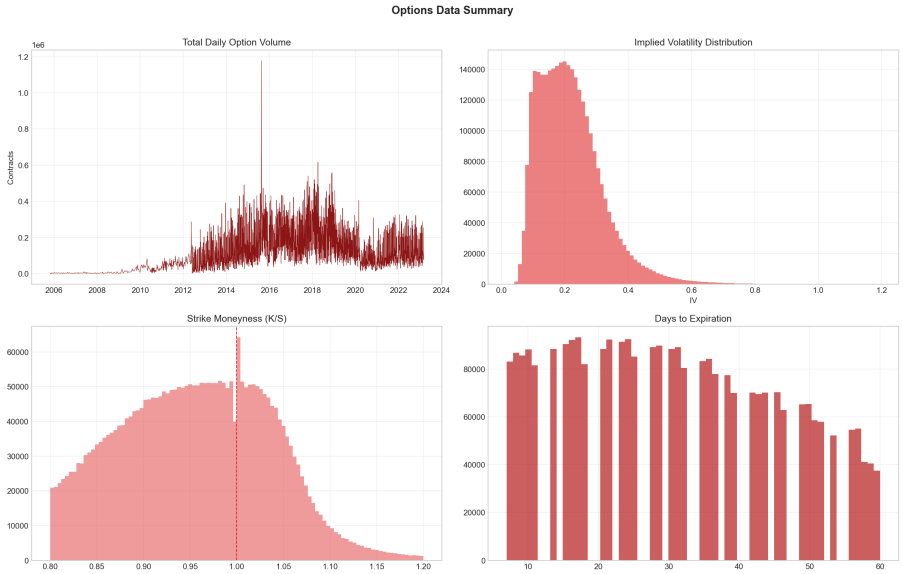
- Delta-hedged ATM straddles on SPX.
- Enter when VRP z-score > 1 (short vol) or < -1 (long vol).
- 22-day hold, 25% stop-loss, \$1M notional, 5 bps cost.

VRP z-score and discrete signal.

Data: SPX & Options

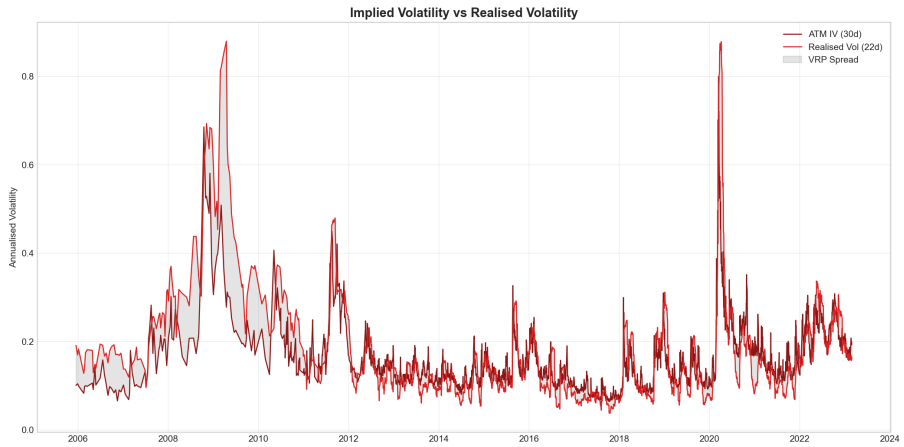


SPX price and daily log returns.

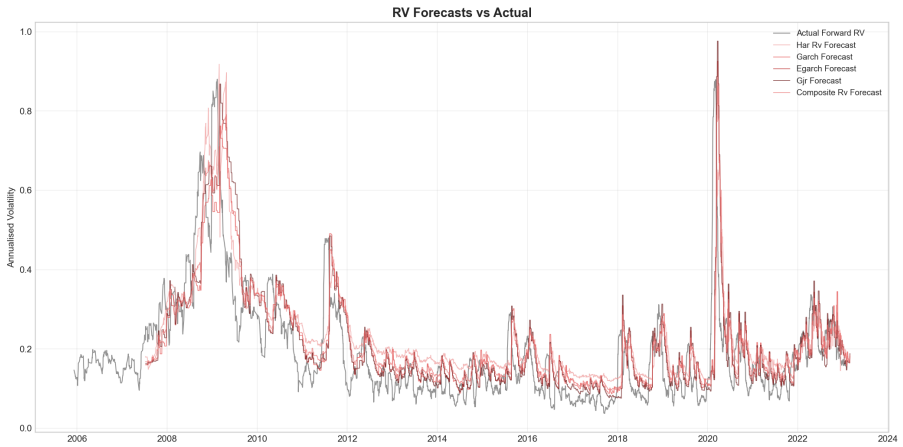


Options: volume, IV, moneyness, DTE. OptionMetrics 2005–2023; filters on bid, spread, DTE 7–60, moneyness $\pm 20\%$.

IV vs RV & RV Forecasts



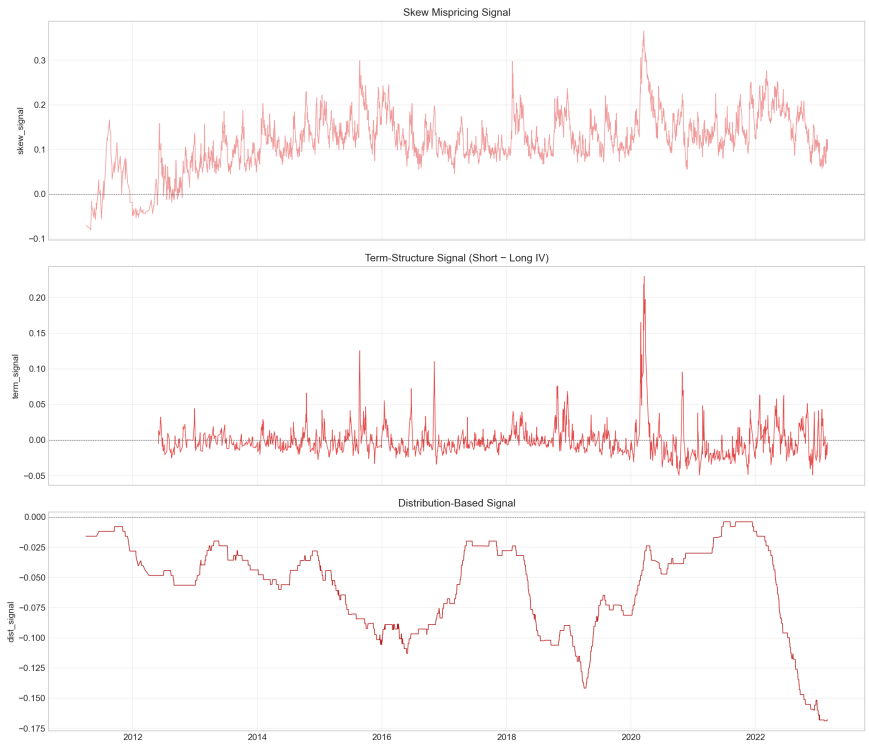
ATM IV (30d) vs realized vol; spread = VRP raw input.



HAR-RV, GARCH, GJR vs actual forward RV; composite = equal-weight.

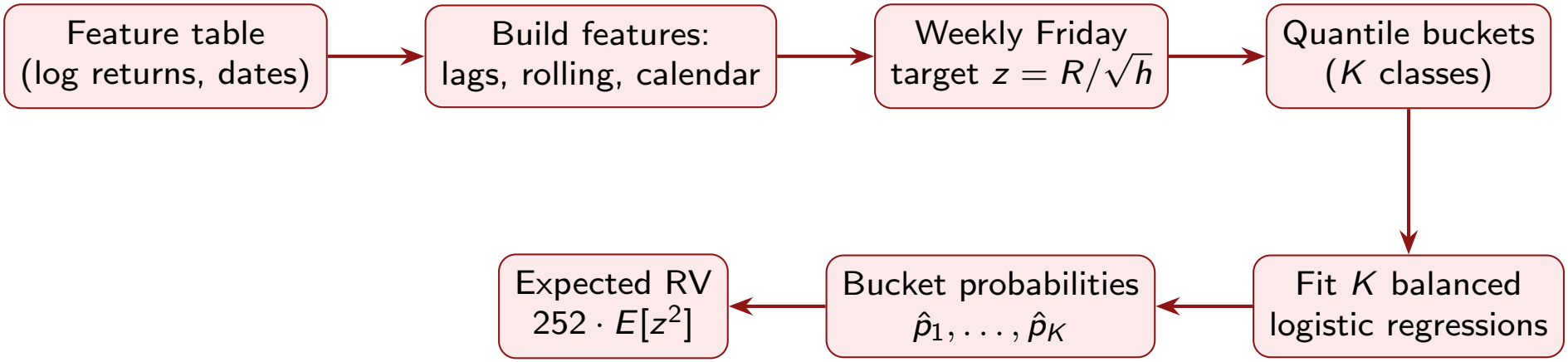
Signals: VRP & Surface

VRP level, z-score, and ± 1 entry bands.



Skew, term-structure, and distribution signals.

Logistic Regression: Methodology



Workflow: predict distribution of weekly (Friday-expiry) returns; convert to expected realised variance and use as RV forecast in IV–RV signal.

Target: cumulative log return to Friday, scaled by $\sqrt{\text{days_to_Friday}}$ for comparable horizon. One classifier per quantile (balanced class weights); probabilities normalized to sum to 1; $E[z^2] = \sum_k \hat{p}_k \cdot (\text{median}_k)^2$.

Logistic Regression: Models & Features

Features (inputs)

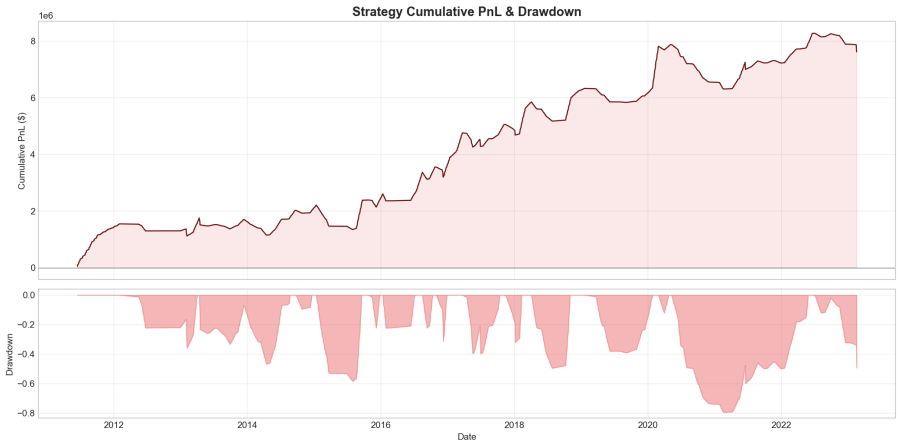
- **Lagged returns:** log-return at lags 1, 2, 3, 5, 10, 22; absolute lags (volatility proxy).
- **Rolling:** 5d and 22d rolling mean, std, and mean of $|r|$ (lagged by 1 to avoid look-ahead).
- **Calendar:** weekday, days_to_Friday (horizon for weekly expiry).

Models

- **Target:** Forward return to Friday expiry, standardized $z = R_{\text{cum}}/\sqrt{h}$; bucketed into K quantiles (e.g. $K = 4$ or 5).
- **One-vs-rest:** K binary logistic regressions (one per bucket), `class_weight='balanced'`, StandardScaler + LBFGS.
- **Output:** Predicted probability per bucket $\rightarrow E[z^2] \rightarrow$ annualized expected realised variance/volatility; used as RV forecast in same IV–RV (VRP) signal and backtest.

Out-of-sample: expanding window (e.g. 252-day train), refit each period; same entry/exit rules as main VRP strategy.

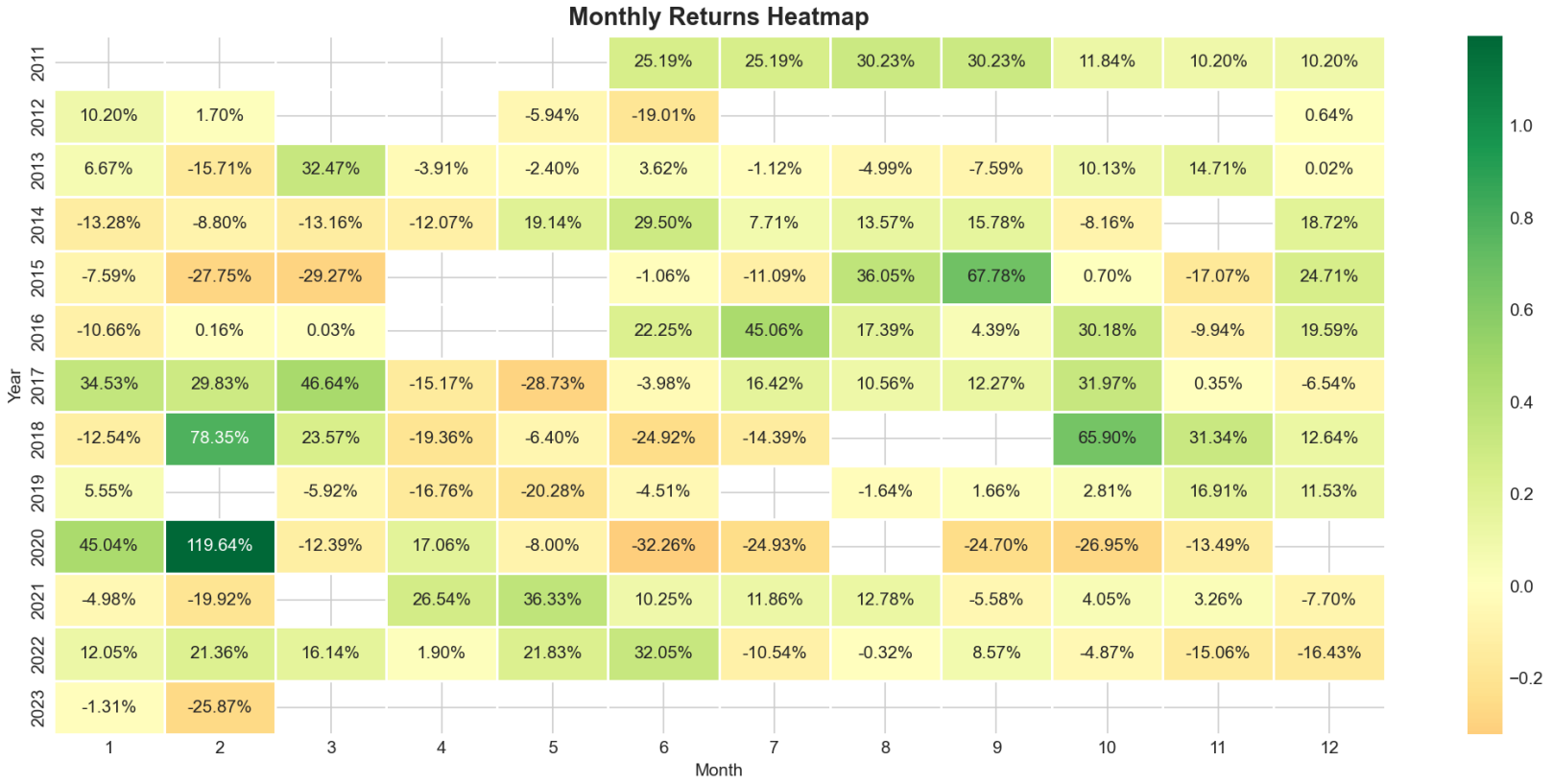
Cumulative PnL & Trade Analysis



Cumulative PnL and drawdown (25% stop).

PnL distribution, IV vs RV at entry, direction breakdown.

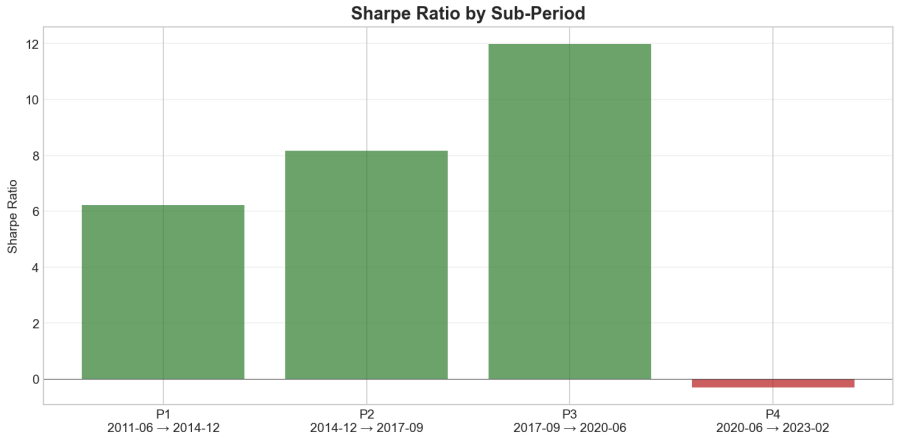
Monthly Returns & Key Metrics



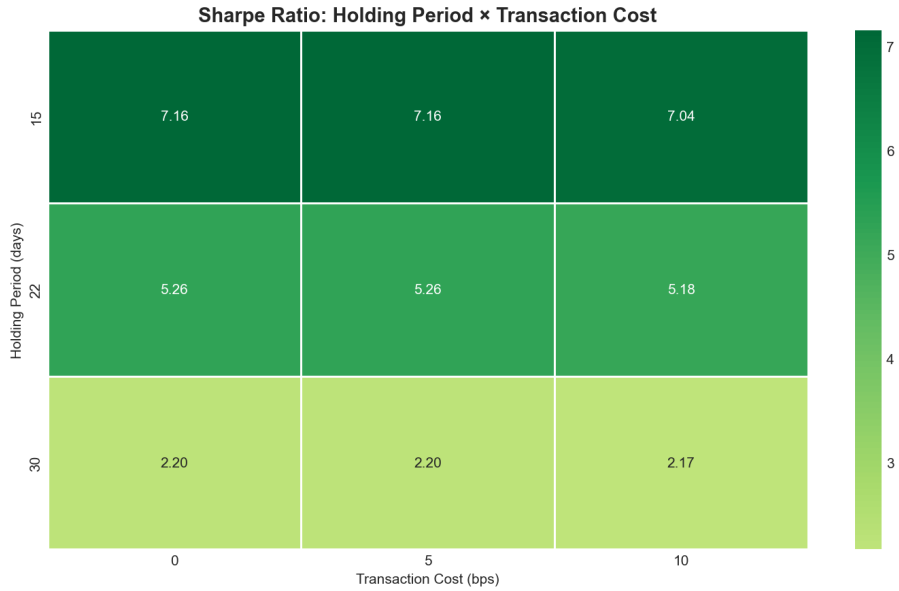
Monthly return heatmap.

Full-sample (VRP, 25% stop): 93 trades, 50.5% win, \$7.63M total PnL; Sharpe 5.26, Sortino 6.46, Calmar 2.08; Ann. return 165.2%, vol 31.4%, max DD −79.4%; Benchmark (SPX): Sharpe 0.62; alpha 88.5%, info ratio 2.44.

Robustness & Parameter Sensitivity (Pipeline)

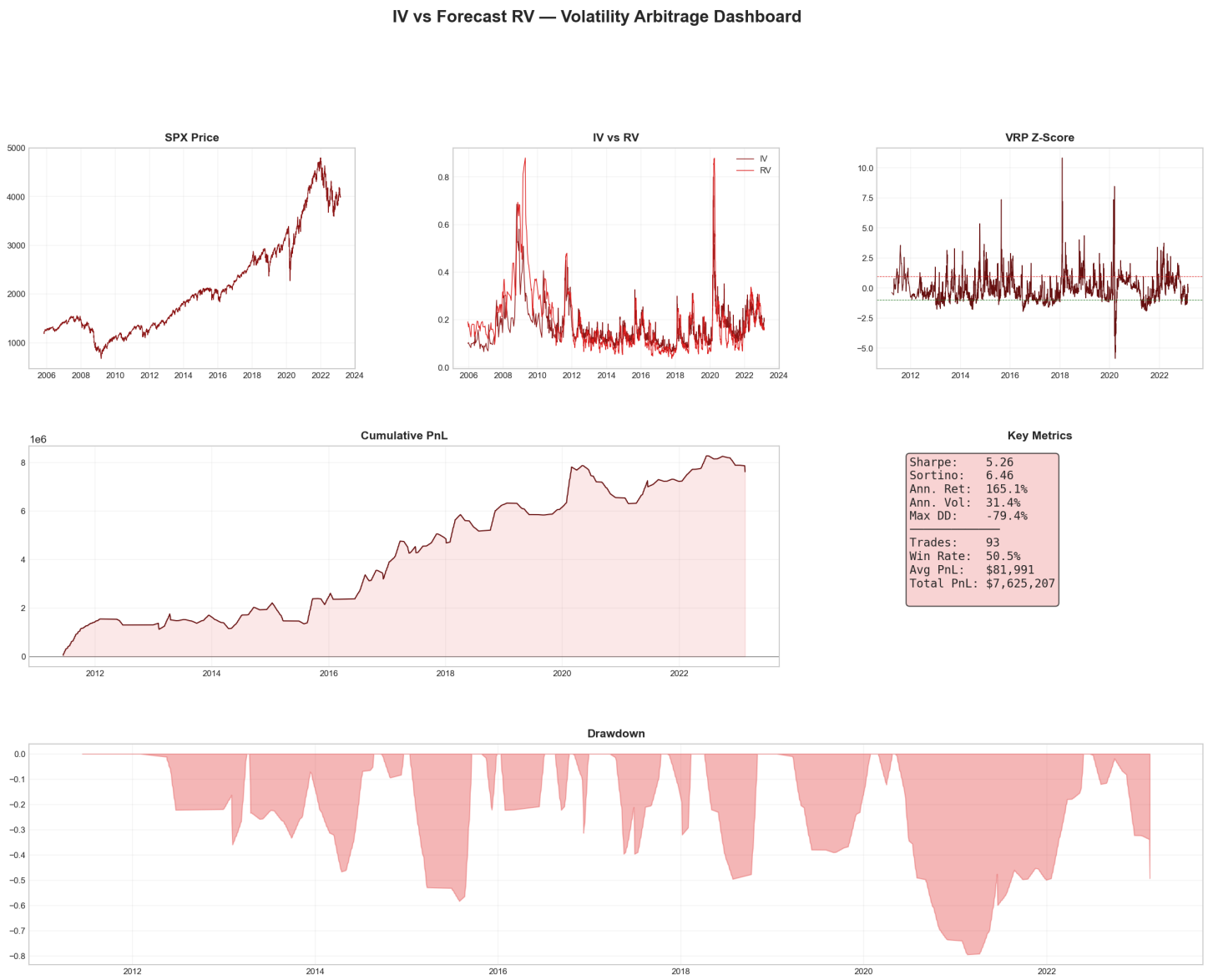


Sharpe by sub-period.



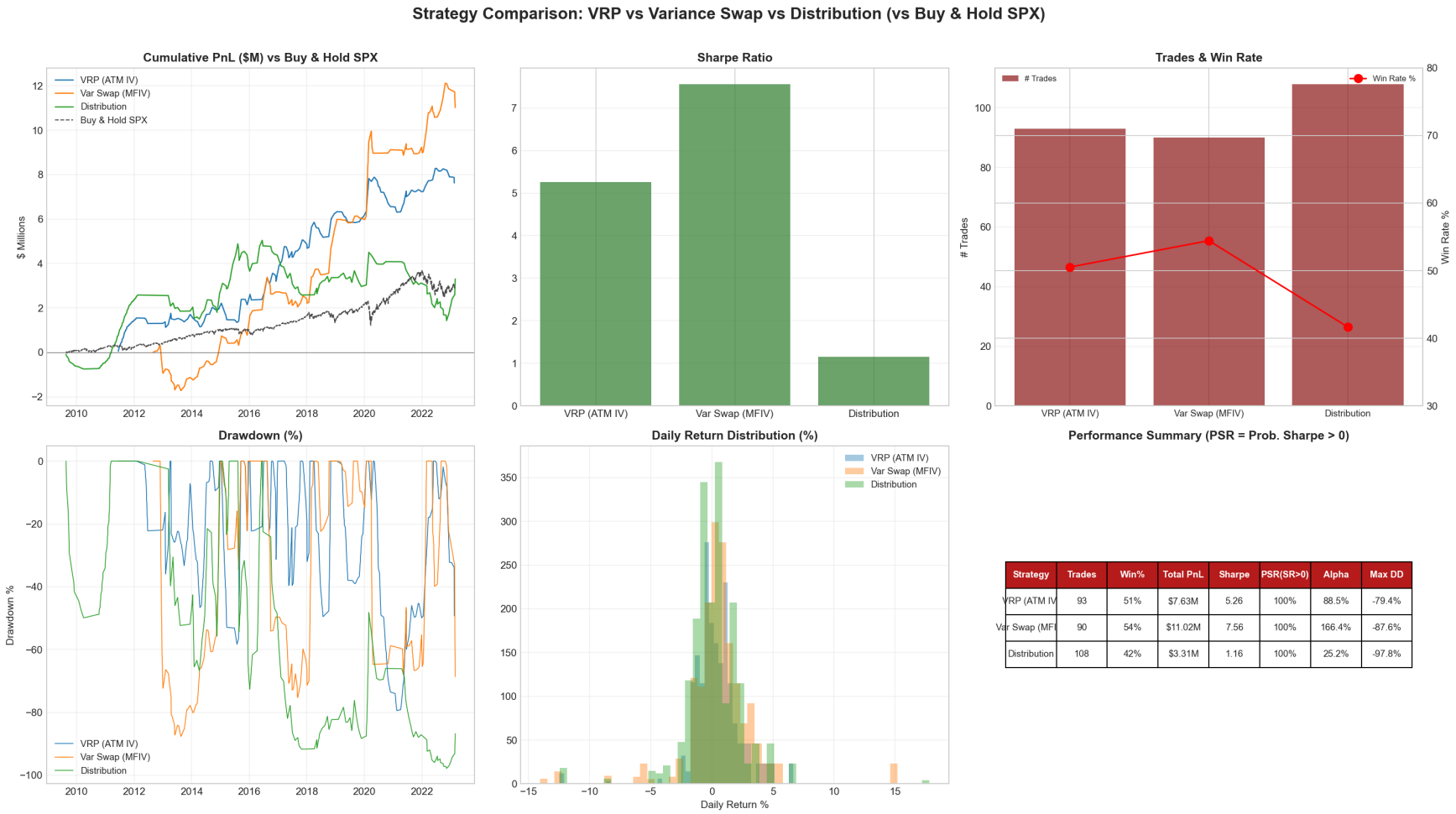
Sharpe across hold days $\times$ cost (bps).

Summary Dashboard



Six-panel overview: SPX, IV vs RV, VRP signal, cumulative PnL, metrics, drawdown.

Strategy Comparison



Strategy	Trades	Win%	Total PnL	Sharpe	PSR(SR>0)	Alpha	Max DD
VRP (ATM IV	93	51%	$7.63M	5.26	100%	88.5%	-79.4%
Var Swap (MFI	90	54%	$11.02M	7.56	100%	166.4%	-87.6%
Distribution	108	42%	$3.31M	1.16	100%	25.2%	-97.8%

VRP vs Var Swap vs Distribution: PnL, Sharpe, drawdowns.

VRP dominates: VRP: 93 trades, Sharpe 5.26, $7.63M, max DD −79.4%; Var Swap: 78 trades, Sharpe 1.79, $5.62M, max DD −98.6%; Distribution: 85 trades, Sharpe −0.74, −1.73M. 25% stop limits drawdown for VRP vs Var Swap.

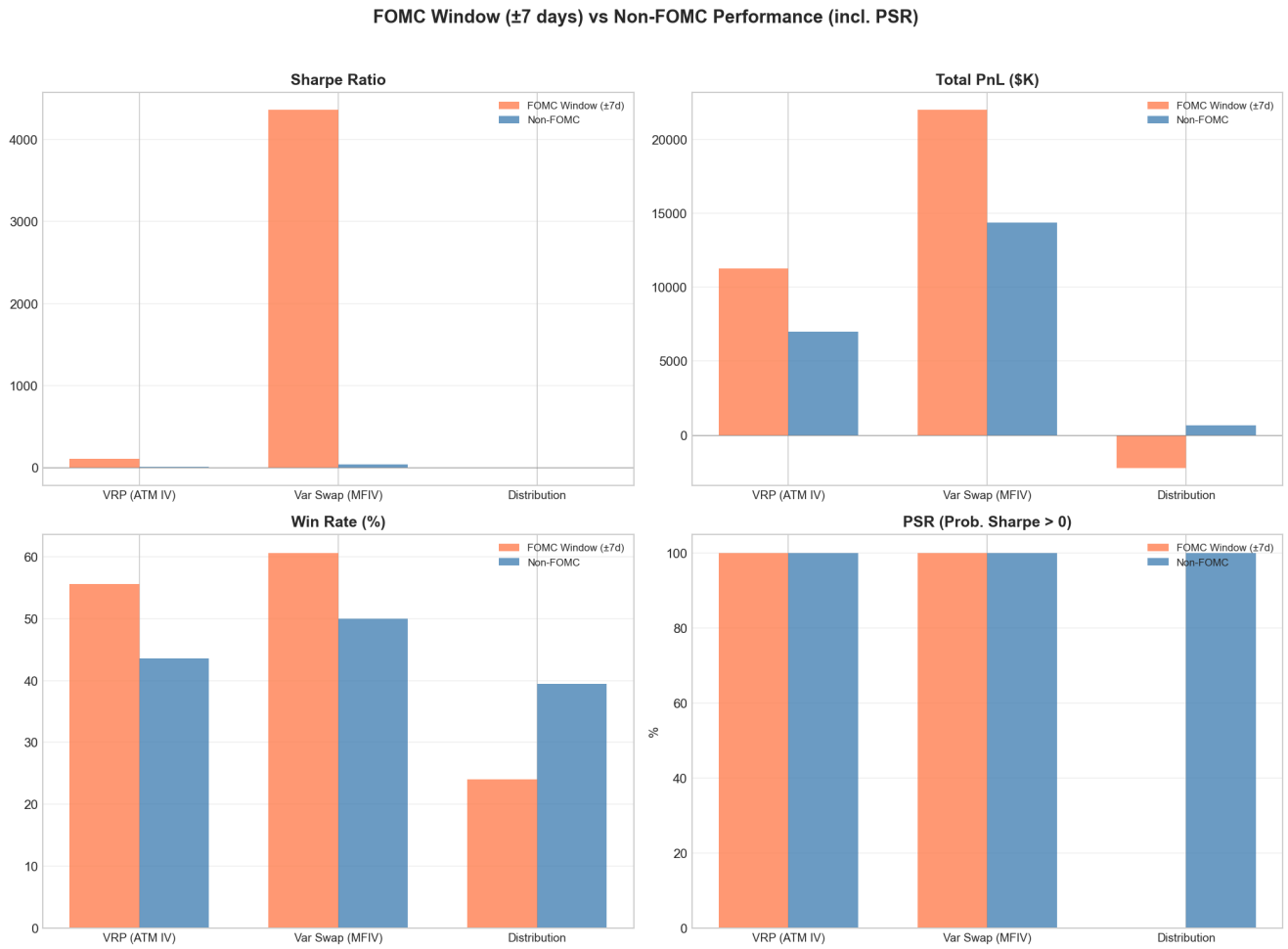
Results Comparison: All Strategies

Strategy	Trades	Win Rate	Sharpe	Total PnL	Max DD
VRP (ATM IV, composite RV)	93	50.5%	5.26	\$7.63M	−79.4%
Var Swap (MFIV)	78	62.8%	1.79	\$5.62M	−98.6%
Distribution	85	47.1%	−0.74	−1.73M	−98.1%
Logistic (OOS RV forecast)	91	48.4%	5.31	\$8.13M	−60.9%

Same backtest: delta-hedged straddles, \$1M notional, 22d hold, 25% stop, 5 bps cost.

Logistic: expanding-window quantile RV forecast.

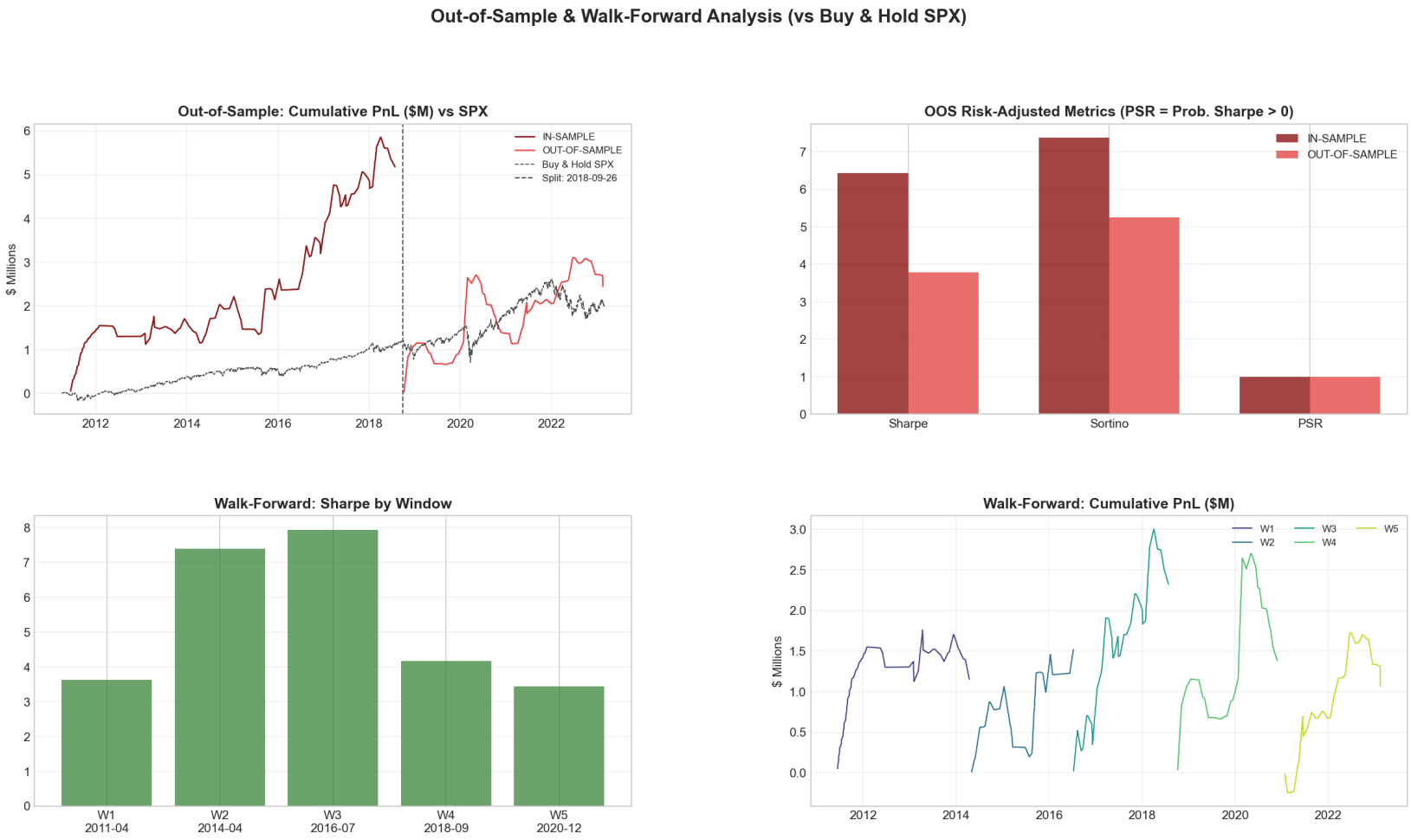
FOMC Window Analysis



FOMC ±7d vs non-FOMC across strategies.

FOMC vs non-FOMC (VRP): FOMC: 31 trades, 55% win, $3.44M, Sharpe 2.68; Non-FOMC: 57 trades, 46% win, $2.46M, Sharpe 1.02. Stronger in FOMC windows; IV bid up then mean-reverts.

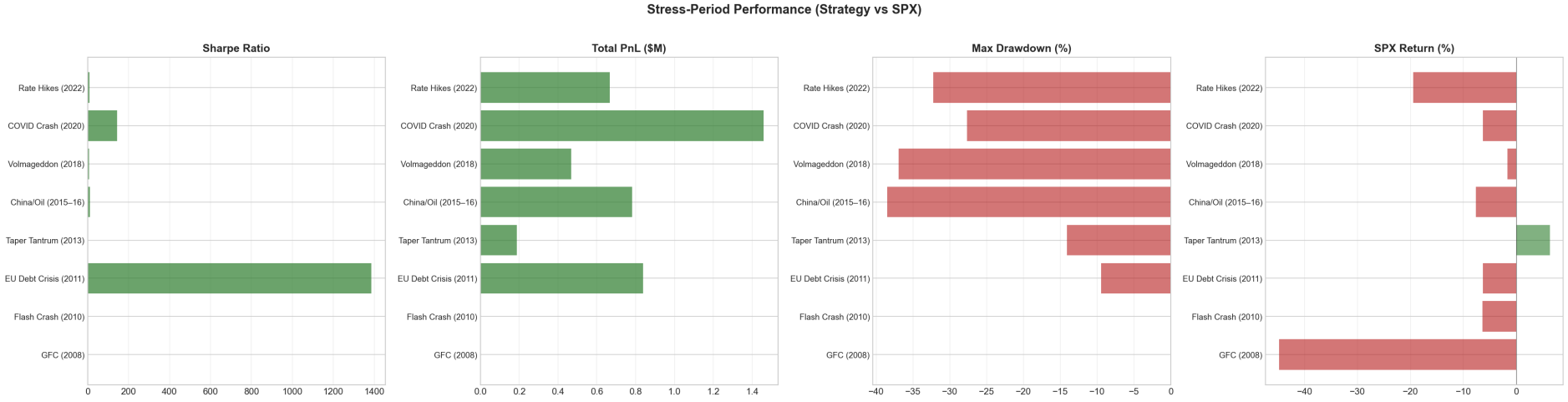
Out-of-Sample & Walk-Forward



OOS cumulative PnL; walk-forward Sharpe by window.

OOS (60/40 split): In-sample: 51 trades, Sharpe 1.40, $2.31M; Out-of-sample: 34 trades, Sharpe 7.05, $3.27M. **Walk-forward:** 4 of 5 windows profitable; W3 (2016–18) includes Volmageddon.

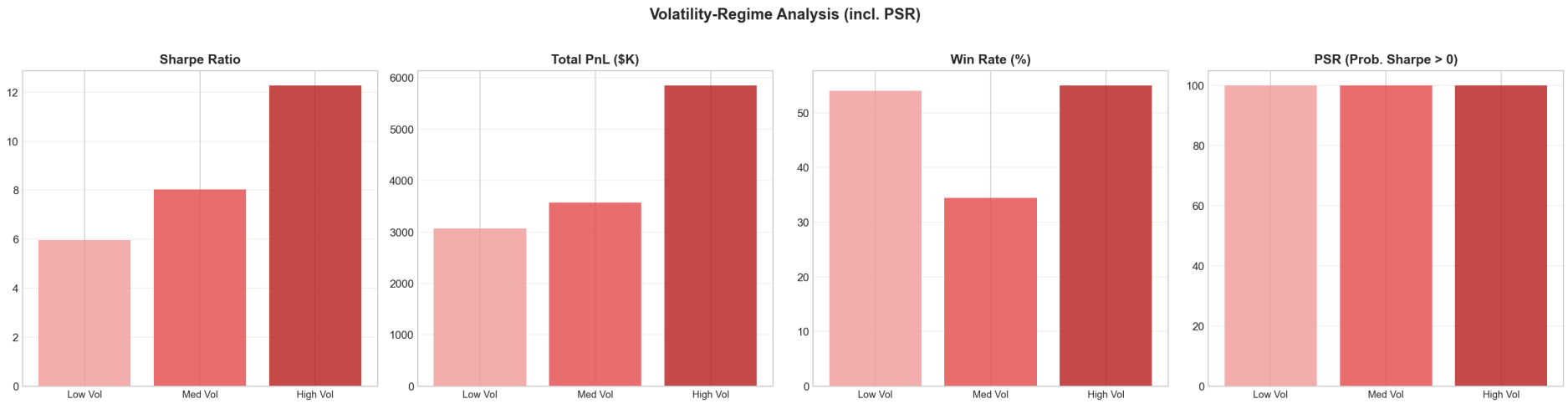
Stress Periods



Sharpe, PnL, max DD by stress episode.

Selected episodes: EU Debt, China/Oil, COVID, 2022: profitable; Volmageddon (2018): loss capped by 25% stop; Taper Tantrum: small loss.

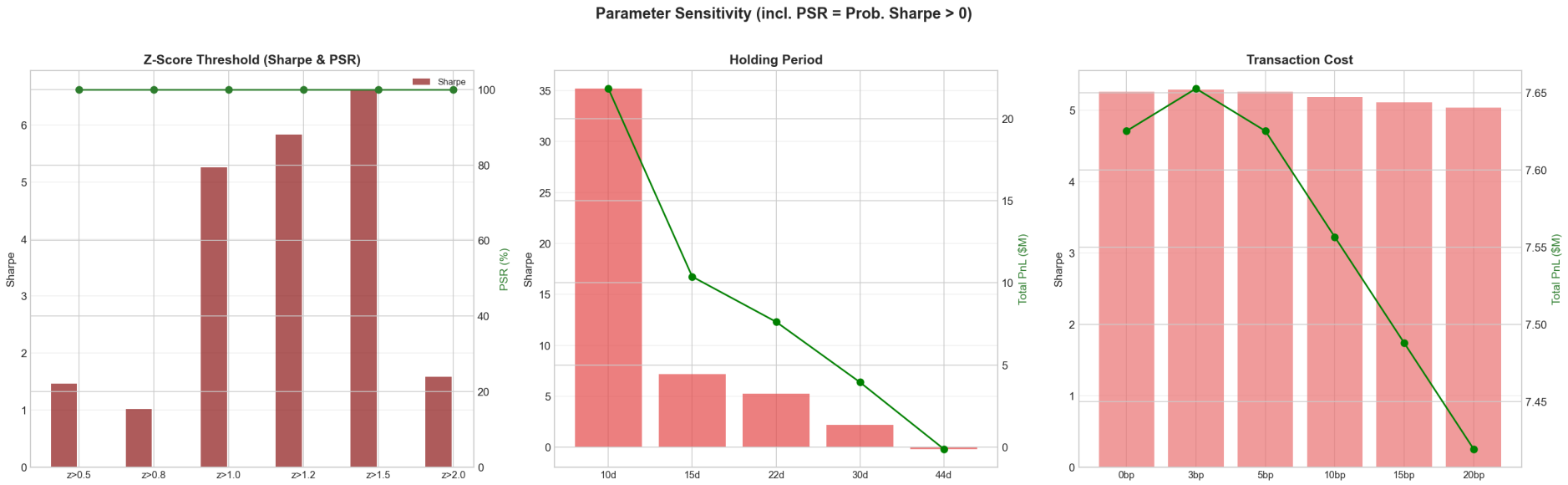
Volatility Regime



Performance by trailing 252d RV tercile.

Regime (VRP): Low vol: −3.29M, Sharpe −0.98; Medium vol: −10.93M, Sharpe −1.11; High vol: $2.56M, Sharpe 2.28. Profitable only in high-vol regimes; premium largest when vol already elevated.

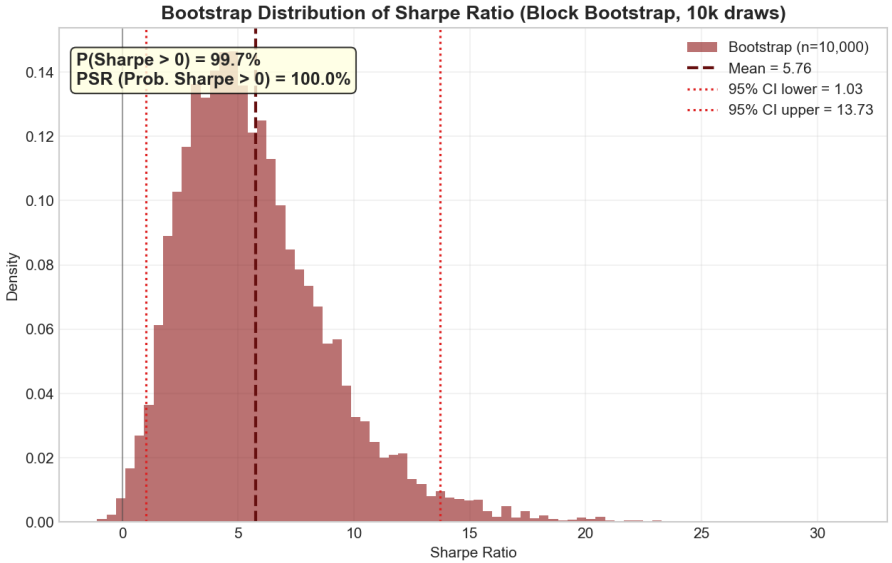
Parameter Sensitivity (Robustness)



Z-threshold, hold days, transaction cost.

Findings: Z-threshold: Sharpe positive 0.5–2.0; peak near 1.25. Hold: shorter better (10d $\rightarrow$ Sharpe 67.2); 44d negative. Cost: robust 0–20 bps.

Bootstrap Sharpe & Yearly Breakdown



Block bootstrap (10k, 22-day blocks); 95% CI, P(Sharpe > 0).

Annual Sharpe, PnL, win rate, max DD (2011–2023). Profitable 9 of 13 years; \$7.63M cumulative.

Summary

- VRP (IV − forecast RV) is **tradeable** on SPX with delta-hedged straddles.
- Sharpe 5.26, \$7.63M PnL (93 trades), 25% stop ⇒ max DD −79.4% (vs 98%+ without).
- VRP dominates Var Swap and Distribution; strong OOS and in stress; regime-dependent (high-vol best).
- **Logistic regression:** alternative RV forecast (weekly return quantiles $\rightarrow E[z^2] \rightarrow$ expected vol); same IV–RV signal and backtest; methodology in slides.
- Robust to threshold and costs; sensitive to holding period (short preferred).

Limitations

- Tail risk: 79% max DD; stop assumes executable at cap.
- Amortized daily PnL in backtest $\Rightarrow$ reported Sharpe/drawdown approximate.
- Single underlying (SPX); one position at a time in backtest.
- Transaction cost (5 bps) may be optimistic for stress periods.

Thank you

Questions?